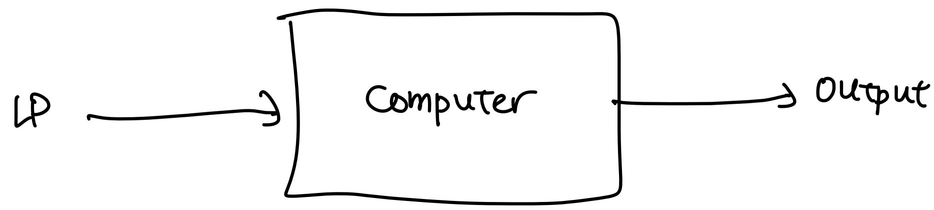
CO 250 Review 3: Certificates of LPs

Module 2: Solving linear programs

Main algorithm: simplex. Efficient to run.

In real life, we ask the computer to run simplex, which outputs a solution.



How do we know that the computer has solved it correctly? We need it to produce a short proof of the result using "certificates". Certificates are "objects" (vectors in our case) that let us verify the result of the computation quickly.

[Recall: In MATH 135, a certificate for verifying $\gcd(a,b)=d$ consists of two integers x, y, and we can quickly verify that $d \mid a$, $d \mid b$ and $ax+by=d$.]

Note: For now, these certificates are given to you. Later in the course, you will learn how to find these certificates.

Outcomes of LPs

What are the possible results of solving an LP?

① It has an optimal solution. $\max x$ s.t. $x \le 3$, $x \ge 0$. $x=3$ is optimal.

② It is infeasible. $\max x$ s.t. $x \ge 3$, $x \le 2$. No x satisfies all constraints.

③ It is unbounded. $\max x$ s.t. $x \ge 3$. There is a series of feasible solutions whose objective values approach ∞. (For min problems, values approach $-\infty$.)

Nice property of LPs: Every LP has one of these 3 outcomes.

Certifying outcomes

Setting: LPs of the form

$$\begin{aligned} \max\ & c^T x \\ \text{s.t. } & Ax = b \\ & x \ge \mathbb{0}. \end{aligned}$$

"Standard equality form"

① Certificate of infeasibility: To prove that no feasible solutions exist, produce a vector y such that $y^T A \ge \mathbb{0}^T$ and $y^T b < 0$. y is the certificate of infeasibility.

Example:

$$\begin{pmatrix} 2 & -1 & 5 & 4 \\ 0 & -1 & 3 & 2 \end{pmatrix} x = \begin{pmatrix} 3 \\ 5 \end{pmatrix}$$
$$x \ge \mathbb{0}$$

Consider $y = \begin{pmatrix} 2 \\ -3 \end{pmatrix}$. Multiply y^T on both sides of the constraints

$$\underbrace{(2 \ \ -3)}_{y^T} \underbrace{\begin{pmatrix} 2 & -1 & 5 & 4 \\ 0 & -1 & 3 & 2 \end{pmatrix}}_{A} x = \underbrace{(2 \ \ -3)}_{y^T} \underbrace{\begin{pmatrix} 3 \\ 5 \end{pmatrix}}_{b}$$

Note: This is 2 times 1st row plus -3 times the 2nd row.

$$\Rightarrow \underbrace{\underbrace{(4 \ \ 1 \ \ 1 \ \ 2)}_{\ge \mathbb{0}^T} \underbrace{x}_{\ge \mathbb{0}}}_{\ge 0} = \underbrace{-9}_{< 0}$$

$\ge \mathbb{0}$ ← vectors

impossible.

≥ 0 ↑ scalar

② Certificate of unboundedness: If there exist $\bar{x}, d$ such that $\bar{x}$ is feasible, $d \geq \mathbb{0}$, $c^T d > 0$ and $Ad = \mathbb{0}$, then the LP is unbounded.

Example:
$$\begin{aligned} \max\ & (2\ 0\ 0)x \\ \text{s.t. } & \begin{pmatrix} -1 & 1 & 0 \\ -2 & 0 & 1 \end{pmatrix} x = \begin{pmatrix} 3 \\ 4 \end{pmatrix} \\ & x \geq \mathbb{0} \end{aligned}$$

Consider $\bar{x} = \begin{pmatrix} 0 \\ 3 \\ 4 \end{pmatrix}$, $d = \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}$.

Consider the solutions $\bar{x} + td = \begin{pmatrix} 0 \\ 3 \\ 4 \end{pmatrix} + t \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}$ for $t \geq 0$.

Feasibility: $A(\bar{x} + td) = A\bar{x} + t(Ad) = \begin{pmatrix} -1 & 1 & 0 \\ -2 & 0 & 1 \end{pmatrix} \begin{pmatrix} 0 \\ 3 \\ 4 \end{pmatrix} + t \begin{pmatrix} -1 & 1 & 0 \\ -2 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}$

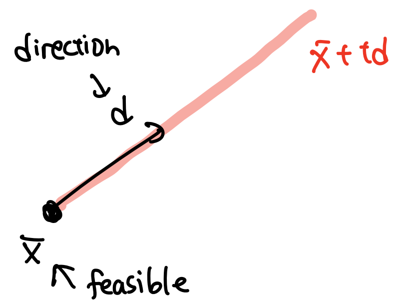
$$= \underbrace{\begin{pmatrix} 3 \\ 4 \end{pmatrix}}_{A\bar{x} = b} + t \underbrace{\begin{pmatrix} 0 \\ 0 \end{pmatrix}}_{Ad = \mathbb{0}} = \begin{pmatrix} 3 \\ 4 \end{pmatrix}$$

$$\bar{x} + td = \underbrace{\begin{pmatrix} 0 \\ 3 \\ 4 \end{pmatrix}}_{\bar{x} \geq \mathbb{0}} + t \underbrace{\begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}}_{d \geq \mathbb{0}} \geq \mathbb{0} \text{ when } t \geq 0.$$

Objective Value: $c^T(\bar{x} + td) = c^T\bar{x} + t(c^T d) = (2\ 0\ 0)\begin{pmatrix} 0 \\ 3 \\ 4 \end{pmatrix} + t\,(2\ 0\ 0)\begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}$

$$= \underbrace{0}_{\text{const}} + \underbrace{2}_{c^T d > 0}t \to \infty \quad \text{as } t \to \infty$$

Geometrically:



As t increases, the points on this ray are feasible, and objective value increases.

③ Certificate of optimality: Prove that $\bar{x} = (2\ 0\ 0\ 4\ 0)^T$ is optimal for

$$\max\ (-1\ \ 3\ \ -5\ \ 2\ \ 1)x - 3$$
$$\text{s.t.} \begin{pmatrix} 1 & -2 & 1 & 0 & 2 \\ 0 & 1 & -1 & 1 & 3 \end{pmatrix} x = \begin{pmatrix} 2 \\ 4 \end{pmatrix}$$
$$x \geq \mathbb{0}$$

The objective value of $\bar{x}$ is 3, and we want to show that every feasible solution has objective value at most 3.

Consider $y = \begin{pmatrix} -1 \\ 2 \end{pmatrix}$. (certificate) Multiply y^T on both sides of the constraints to get

$$(-1\ \ 2)\begin{pmatrix} 1 & -2 & 1 & 0 & 2 \\ 0 & 1 & -1 & 1 & 3 \end{pmatrix} x = (-1\ \ 2)\begin{pmatrix} 2 \\ 4 \end{pmatrix} \Rightarrow (-1\ \ 4\ \ -3\ \ 2\ \ 4)x = 6.$$

Rewrite this as $0 = -(-1\ \ 4\ \ -3\ \ 2\ \ 4)x + 6$.

We can add "0" to the objective function without changing anything.

Now objective: $(-1\ \ 3\ \ -5\ \ 2\ \ 1)x - 3 + \big(-(-1\ \ 4\ \ -3\ \ 2\ \ 4)x + 6\big)$

$$= \underbrace{\underbrace{(0\ \ -1\ \ -2\ \ 0\ \ -3)}_{\leq 0^T}\ \underbrace{x}_{\geq 0}}_{\leq 0} + 3$$

So any feasible solution has value at most 3. Hence $\bar{x}$ is optimal.